MarketSage AI

Master Engineering Blueprint — **Part VI**

Dashboard and Phase-Gated Roadmap

The product surface, alerting, and the ten-phase execution plan with numeric exit gates, kill criteria and the first fourteen days.

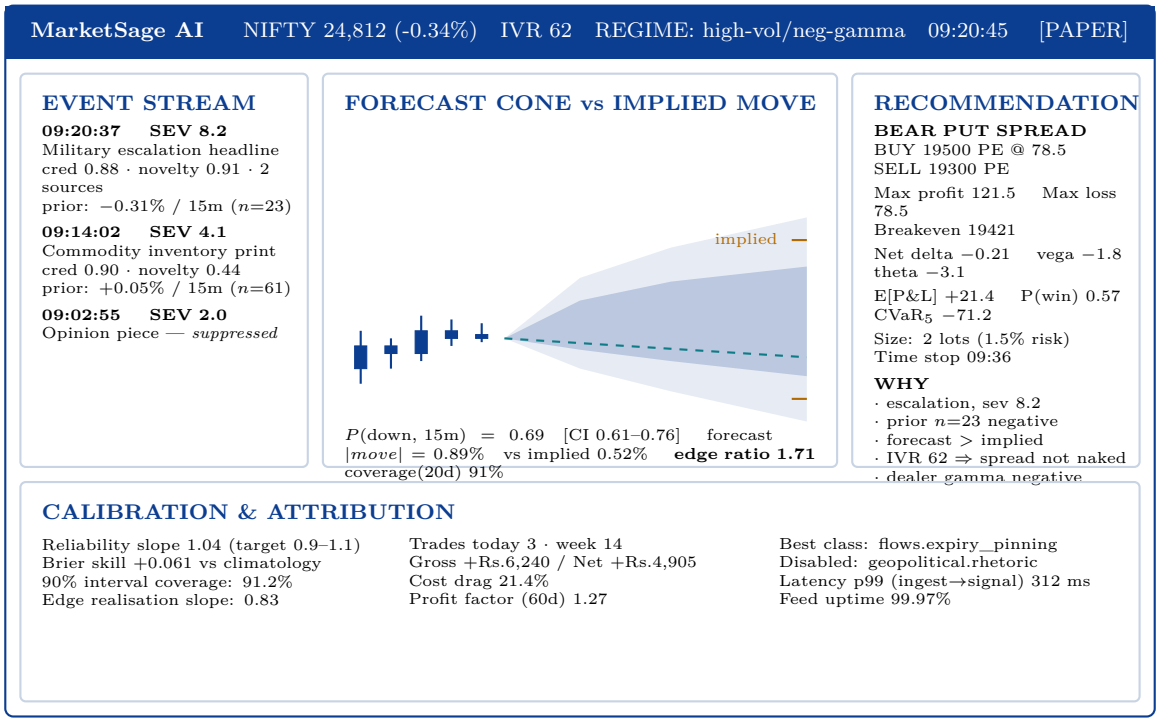
This is an extract of the MarketSage AI master plan, published separately for convenience. Chapter, section and equation numbering matches the full document, and cross-references point back into it. Read Chapter 3 (Reality Calibration) first — every design decision in this part follows from it.

CHAPTER 16

Dashboard and Product Surface

16.1 Design brief

The dashboard has one job: let a human decide in three seconds whether to act, and let the same human audit six weeks later why the machine acted. Everything else is decoration.



16.2 Panels and their non-negotiable elements

Panel	Must contain
Event stream	Severity, credibility, novelty, source count, the historical prior *with its sample size*, and suppressed events shown greyed out so the user sees what the system chose to ignore.
Forecast cone	The quantile bands, the median as a thin dashed line, the implied-move band for comparison, the edge ratio, and the trailing coverage statistic so the user knows whether to trust the bands today.
Recommendation	The full structure with both legs, net Greeks, the Monte Carlo summary, the sizing rationale, all three exits (target, stop, time), and the ordered "why" list.
Calibration & attribution	Always visible, never hidden behind a tab. The system's honesty is the product.
Status bar	Mode (PAPER/LIVE/DEGRADED), regime, latency, feed health. Mode must be impossible to misread — a full-width coloured band when live.

Interaction rules

- Nothing flashes or animates except a genuinely new high-severity event.
- Every number is hoverable to reveal its computation and its as-of timestamp.
- Any recommendation can be opened into a full replay: the news item, the feature vector, the model versions, the Monte Carlo distribution.
- A permanent, unmissable indicator of PAPER versus LIVE mode.

16.3 Alerting

Alert	Channel	Rule
Tradeable signal	Push + sound	Only when all gates pass. Rate-limited to avoid alert fatigue.
High-severity event, no trade	Push, silent	"Something is happening, risk is elevated" — valuable even without a trade.
Position exit	Push	Target, stop or time stop hit.
Risk breach	Push + SMS	Any L2/L3 kill switch.
System health	Email/chat	Daily digest; immediate only for L3.
Calibration decay	Daily digest	Slow-moving; does not need to interrupt.

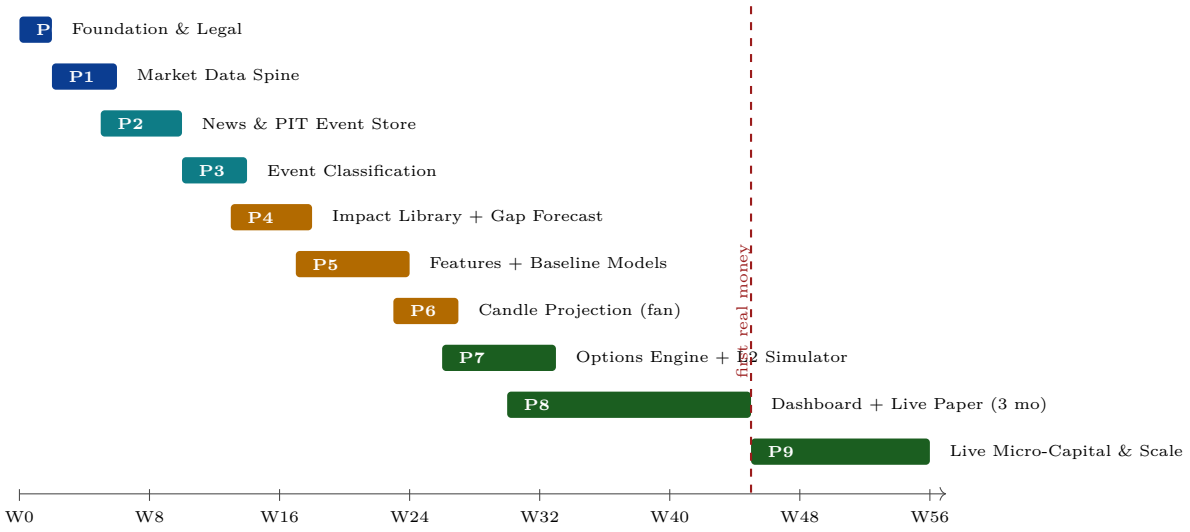
CHAPTER 17

Phase-Gated Execution Roadmap

17.1 How to read this chapter

Ten phases. Each has an objective, workstreams, checkpoints, a numeric **exit gate**, and **kill criteria**. You may not begin phase $n+1$ until phase n's gate is signed off in writing (a dated entry in `docs/gates.md` with the measured numbers pasted in).

The durations assume one committed developer working roughly 25–30 focused hours per week. With a second person on data engineering, compress by around 30%. Do not compress by skipping gates.



17.2 Phase 0 — Foundation and Legal (Weeks 1–2)

Objective. Make it impossible to build the wrong thing, and start the data clock.

CP	Checkpoint	Definition of done
0.1	Repository and environment	Monorepo scaffolded per Chapter 18, Docker Compose brings up Postgres/TimescaleDB, Redis and a service skeleton with one command. Pre-commit hooks including a secret scanner.
0.2	Broker and data vendor decision	Written comparison of at least three brokers on: WebSocket instrument limits, tick frequency, historical data depth, option chain access, algo registration process, order API reliability. Decision recorded with reasons.

0.3	Compliance memo	One page: which regulations apply to personal automated trading through the chosen broker, what data may be stored, what may be shown to others, and what changes if a second user is added. **Written before architecture is frozen.**
0.4	Recorder running	A minimal always-on process writing ticks, option chain snapshots and news headlines with timestamps to disk. Crude is fine. It runs tonight.
0.5	Instrument master	Canonical instrument table, expiry calendar, holiday calendar, lot sizes, tick sizes — all loaded and versioned.

Gate G0

The recorder has produced 5 consecutive complete trading days of ticks, chain snapshots and news with zero unexplained gaps, and the compliance memo is signed. **Every day this gate slips is a day of irrecoverable data.**

17.3 Phase 1 — Market Data Spine (Weeks 3–6)

Objective. A correct, replayable, queryable market history.

CP	Checkpoint	Definition of done
1.1	Tick ingestion service	Reconnect logic, sequence-gap detection, backpressure handling, heartbeat monitoring. Survives a forced disconnect mid-session without data loss.
1.2	Candle builder	1/5/15-minute candles per Chapter 5 rules, including synthetic gap candles and session tagging.
1.3	Option chain snapshotter	Both indices, near two expiries, full strike range, at 1–5 second cadence with completeness metrics.
1.4	TimescaleDB schema + compression + retention	Hypertables, continuous aggregates, compression policy, and a tested restore from backup.
1.5	Parquet archival + DuckDB research layer	A notebook can scan a month of candles in under a second.
1.6	Replay engine	Deterministic replay of a recorded day through the bus. Byte-identical candle output on repeat runs (automated test).
1.7	Data quality gates	The checks in Chapter 5 running nightly with alerts.

Gate G1

(i) Replay a chosen historical day and reproduce candles byte-identically twice. (ii) 20 consecutive sessions captured with $\geq 99.5\%$ expected candles present and option chain completeness $\geq 95\%$. (iii) A full restore from backup succeeds on a clean machine.

Kill criteria for P1

If the chosen broker feed cannot deliver adequate option chain coverage, stop and change vendor now. Discovering this at Phase 7 wastes four months.

17.4 Phase 2 — News Ingestion and Point-in-Time Event Store (Weeks 6–10)

Objective. Build the asset nobody else has: a timestamp-honest archive of what was knowable when.

CP	Checkpoint	Definition of done
2.1	Collector framework	Pluggable per-source collectors, per-source rate limits, retries with backoff, health heartbeats, and a source registry with credibility weights.
2.2	Publish-time resolution	Extract the true publication timestamp; where impossible, set `ts_uncertain` and record the estimation method. Measured accuracy against a manually verified sample.
2.3	Deduplication	Exact hash, SimHash and embedding ANN with cluster heads. Measured duplicate rate.
2.4	Economic calendar with consensus	Scheduled events with expected values loaded ahead of time; surprise computable at release.
2.5	Historical backfill	As much archive as licensing permits, clearly separated from the live-collected archive because their timestamp quality differs.
2.6	Storage and search	PostgreSQL plus vector index; "show me every event of class X in the last two years" returns in seconds.

Gate G2

30 consecutive days collected. Duplicate rate below 3%. On a manually verified sample of 200 items, the publish timestamp is within 5 seconds of truth for at least 90% of items. A point-in-time query ("what did we know at 2026-07-14 11:03:00?") returns a correct, reproducible set.

17.5 Phase 3 — Event Classification (Weeks 10–14)

Objective. Turn text into structured, scored events.

CP	Checkpoint	Definition of done
3.1	Taxonomy v1 frozen	Written definitions with positive and negative examples for every class, including the noise class.

3.2	Gold labelled set	2,000–3,000 items labelled, two annotators on at least 20%, inter-annotator agreement measured and above an agreed threshold. Use an LLM to pre-label and humans to correct — roughly a five-fold speed-up.
3.3	Classifier trained	Fine-tuned encoder, multi-label head, severity regression head, temperature-calibrated confidence.
3.4	Entity and geo extraction	Linked to the canonical knowledge graph, not free text.
3.5	Severity empirical recalibration	Section 6.3 step 3 completed against realised abnormal moves.
3.6	Novelty and confirmation logic	Implemented with the fake-news defence tests from Chapter 6.
3.7	Latency optimisation	Quantised or ONNX-exported; p95 under 120 ms for the tier-2 model.
3.8	Two-tier funnel wired into the bus	End-to-end from raw item to `event.classified`.

Gate G3

Macro-F1 ≥ 0.80 on a temporally held-out test set. Recall on severity-$\geq$7 events ≥ 0.95. False-positive rate on the noise class $\leq 2\%$. Tier-2 p95 latency ≤ 120 ms. The AP-hack injection test produces no directional signal. Severity correlates with realised absolute abnormal return with Spearman $\rho \geq 0.35$.

The severity correlation is the real gate

A classifier can score beautifully on labels and be useless for trading. If severity does not correlate with realised market movement, the taxonomy or the labelling rubric is wrong — fix that before proceeding. This is the most common silent failure in the entire project.

17.6 Phase 4 — Impact Library and the Gap Forecaster (Weeks 13–18)

Objective. Learn what events historically did, and ship the first genuinely useful product.

CP	Checkpoint	Definition of done
4.1	Abnormal-return engine	Time-of-day and regime normalised abnormal returns with matched control sampling.
4.2	Event study over the full archive	Every historical event scored across all horizons; results stored and queryable.
4.3	Hierarchical Bayesian posteriors	Per class and family, with shrinkage, credible intervals and effective sample sizes.
4.4	Volatility and implied-volatility response library	The $\sigma_{\text{post}}/\sigma_{\text{pre}}$ and $\Delta\sigma_{IV}$ distributions per class.
4.5	Cross-asset transfer coefficients	Estimated lead-lag relationships with confidence intervals.

4.6	**Overnight gap forecaster**	A model predicting the next session's opening gap from overnight global events and cross-asset moves — the first component that produces standalone value with no latency requirements.
4.7	Morning briefing product	An automated pre-market note: overnight events, gap forecast distribution, key levels, scheduled events today.

Gate G4

At least 15 event classes have posteriors with $n_{\text{eff}} \geq 8$ and credible intervals excluding zero for at least 5 of them. The gap forecaster beats a naive "gap = overnight global futures move" baseline on out-of-sample CRPS by $\geq 10\%$. The morning briefing has been generated automatically for 20 consecutive sessions.

Why the gap forecaster is placed here

It is the fastest path from "interesting research" to "a thing that is useful every single morning". It needs no low-latency infrastructure, no options engine and no live execution. Shipping it at week 18 rather than at week 45 sustains motivation and produces real feedback — and, being a distributional forecast scored by CRPS, it validates the entire probabilistic approach cheaply.

17.7 Phase 5 — Feature Store and Baseline Prediction (Weeks 17–24)

Objective. Find out whether an intraday edge exists at all.

CP	Checkpoint	Definition of done
5.1	Feature framework	The one-definition/two-materialisation pattern with an automated offline–online parity test in CI.
5.2	Feature groups A, C, E, F implemented	Group B added only if the feed supports it.
5.3	Leakage linter	Automated check of every feature's dependency timestamps against the decision time. Fails the build on violation.
5.4	Triple-barrier labelling	With volatility-scaled barriers, sample uniqueness weights and event grouping.
5.5	Purged/embargoed CV harness	Plus combinatorial purged CV producing a Sharpe *distribution*.
5.6	M0 rules baseline	The bar every model must clear.
5.7	M1 LightGBM direction, M5 volatility, M8 regime	Trained, tuned, SHAP analysed.
5.8	Calibration layer	Isotonic fit plus reliability diagrams.
5.9	M7 meta-labeller	Trained on M1 outputs.
5.10	**Historical option chain purchased**	The hard dependency for Phase 7. Order it at the start of this phase; delivery and cleaning take weeks.

Gate G5 — the decision gate of the whole project

On walk-forward out-of-sample data, restricted to event-conditioned samples and with a 3-second decision delay applied:

- M1 beats M0 on log loss by a statistically significant margin (bootstrap CI excludes zero).
- Calibrated: reliability slope in $[0.85, 1.15]$, Brier skill score > 0.02 versus climatology.
- Directional AUC ≥ 0.55 on high-severity event samples.
- M5 volatility forecast beats a HAR-RV baseline on QLIKE.
- Edge realisation slope > 0.5 in the L1 evaluation.

Kill criteria for P5 — read this twice

If, after honest purged cross-validation with the decision delay applied, AUC sits at 0.50–0.52 and calibration is flat, **there is no intraday directional edge in your current feature set**. Do not proceed to build an options engine on top of a coin flip. Instead: (a) pivot to *volatility-only* strategies, where the gate is far easier to pass; (b) pivot to the overnight gap product from Phase 4, which already works; (c) invest in better data (order book depth, faster sources) and retry once. Three failed attempts at G5 means the honest answer is that this specific edge does not exist at your data tier — and knowing that at week 24 instead of week 80 is a success, not a failure.

17.8 Phase 6 — Distributional Candle Projection (Weeks 23–27)

CP	Checkpoint	Definition of done
6.1	M2 quantile models	Pinball-loss heads at each horizon.
6.2	Conformal wrapper	Split conformal intervals with verified coverage.
6.3	Path sampler	Volatility-clustering simulator calibrated so terminal quantiles match M2.
6.4	Fan-chart renderer	Per-minute quantile candles plus the implied-move overlay.
6.5	Projection scoring	Pinball, CRPS, PIT and coverage tracked daily.
6.6	M3 sequence model (optional)	Only if it beats M1 in the ensemble; drop it without sentiment if it does not.

Gate G6

Out-of-sample 90% interval coverage within $90\% \pm 3\%$; PIT histogram passes a uniformity test; CRPS beats both a random-walk-with-GARCH baseline and the option-implied distribution by $\geq 8\%$ relative. Rendering latency under 120 ms for 5000 paths.

17.9 Phase 7 — Options Engine and Simulator (Weeks 26–33)

CP	Checkpoint	Definition of done

7.1	IV surface fitting	Arbitrage-free parametric fit per expiry with quality metrics and stale-quote filtering.
7.2	Greeks engine	Validated against an independent reference implementation to within tolerance.
7.3	M6 implied-volatility change model	Trained and calibrated.
7.4	Structure enumerator	With liquidity eligibility constraints.
7.5	Monte Carlo P&L engine	Vectorised, under 90 ms p99 for the full candidate set.
7.6	Cost model	Full statutory and brokerage stack, point-in-time rates, plus an empirical slippage model.
7.7	Sizing and risk gate	Fractional Kelly with all modifiers and every limit from Chapter 12.
7.8	L2 event-driven simulator	Production code path, injected latency, realistic fills, conservative tie-breaking.
7.9	Stress-test suite	All ten scenarios from Chapter 12 as automated replay tests.
7.10	**Locked-holdout evaluation**	The single permitted evaluation against the untouched 3-month holdout.

Gate G7

On the L2 simulator over at least 12 months including the locked holdout, and after full costs and a 3-second delay: profit factor ≥ 1.2; positive expectancy in at least 3 of 4 quarters; max drawdown $\leq 12\%$ of notional risk capital; deflated Sharpe ratio > 0; and the strategy survives all ten stress scenarios with the specified behaviour. Every recommendation carries a complete explanation record.

17.10 Phase 8 — Dashboard and Three Months of Live Paper Trading (Weeks 30–45)

Objective. Prove it in the wild, with no money at risk. This phase cannot be compressed — it is bounded by calendar time, not by effort.

CP	Checkpoint	Definition of done
8.1	Dashboard v1	All five panels from Chapter 16, WebSocket live, with the PAPER/LIVE indicator.
8.2	Alerting	Push notifications with rate limiting.
8.3	Paper ledger	Every signal recorded with simulated fills at live prices, plus outcome resolution at horizon.
8.4	Daily operating rhythm automated	Preflight, session, post-session, nightly and weekend jobs all running unattended.
8.5	Observability	Trace propagation, dashboards, alarms, error budget.
8.6	**60 trading sessions of live paper**	Unattended, with an availability target of 99% during market hours.
8.7	Weekly attribution reviews	12 written reviews with actions taken.

8.8	Class-level kill policy active	Underperforming event classes automatically disabled.

Gate G8 — the go-live gate

Over ≥ 60 sessions and ≥ 150 paper trades:

- Positive net expectancy after full modelled costs.
- Profit factor ≥ 1.15.
- Max drawdown $\leq 10\%$ of notional risk capital.
- Reliability slope in $[0.85, 1.15]$ on live data.
- Live results within one standard error of the L2 simulator's prediction for the same period (**if live and simulated diverge, the simulator is wrong and everything built on it is suspect**).
- Zero unhandled exceptions in the trading path; every kill switch exercised at least once in a drill.
- Broker algo registration complete; compliance memo updated.

17.11 Phase 9 — Live Micro-Capital and Scaling (Week 45 onward)

Rung	Capital	Promotion criteria to the next rung
R1	Minimum viable (1 lot, smallest instrument)	20 sessions, live P&L within one standard error of paper P&L for the same signals, slippage within the modelled band.
R2	2x R1	40 sessions, profit factor ≥ 1.15, no L3 kill events.
R3	4x R1	60 sessions, Sharpe > 1 annualised, drawdown within limits.
R4	8x R1	Two consecutive profitable quarters; slippage still within model.

Demotion rule: any breach of the weekly loss limit, or live results deviating from paper by more than two standard errors, drops the ladder by one rung immediately and triggers a written review. Scaling up is slow; scaling down is instant.

The paper-versus-live divergence test

Run the paper engine *alongside* the live engine forever, on identical signals. The difference between paper and live P&L is your true execution cost. If that gap widens, the model has not degraded — your fills have. These are different problems with different fixes, and without running both you cannot tell them apart.

17.12 Phase summary table

Ph	Name	Weeks	Single most important output
P0	Foundation and Legal	1–2	The recorder is running tonight.
P1	Market Data Spine	3–6	Deterministic replay.

P2	News and PIT Event Store	6–10	Timestamp-honest archive.
P3	Event Classification	10–14	Severity that correlates with real moves.
P4	Impact Library + Gap Forecaster	13–18	Conditional base rates, and the first shipped product.
P5	Features + Baseline Models	17–24	The honest answer to "is there an edge?"
P6	Candle Projection	23–27	Calibrated forecast cones.
P7	Options Engine + Simulator	26–33	Trades whose expected value is positive after costs.
P8	Dashboard + Live Paper	30–45	60 sessions of unattended live evidence.
P9	Live Capital	45+	A capital ladder that only climbs on evidence.

17.13 The first fourteen days, concretely

Day	Task
1	Create the repository and the Docker Compose stack (Postgres/TimescaleDB, Redis). Get `docker compose up` working. Add pre-commit with a secret scanner.
2	Open the broker account with API access; obtain and store credentials in a secrets manager. Read the API documentation end to end and write down the instrument subscription limit and tick frequency.
3	Write the tick recorder. Subscribe to the two index spots and futures. Write raw messages to disk as newline-delimited JSON. Ugly is fine.
4	Write the option chain snapshotter. Both indices, near two expiries. Run it for a full session and measure completeness.
5	Write two news collectors (one wire RSS, one official/central-bank page). Store raw with three timestamps.
6	Build the instrument master, expiry calendar and holiday calendar. Load them.
7	Review the week: verify that five days of data exist and are complete. Fix gaps. **This recorder never stops again.**
8	TimescaleDB schema; a loader from raw files into hypertables.
9	The candle builder with the six correctness rules; write the determinism test.
10	Parquet archival job and a DuckDB notebook; plot a day of Nifty from Parquet.
11	Data quality checks: candle counts, tick gaps, chain completeness. Wire an alert.
12	The compliance memo. Talk to the broker about algo registration requirements. Write it down.
13	The broker/vendor comparison document; commit to a decision.
14	Gate G0 review. Write the numbers into `docs/gates.md`. Only then start Phase 1 properly.

What not to do in the first fourteen days

Do not train a model. Do not build a dashboard. Do not integrate an LLM. Do not optimise latency. Every one of those is more fun than building a recorder, and every one of them is

worthless without the data the recorder produces.